Example 2.42

Find the 5th term in the expansion of $\left(x-\frac{3}{x^2}\right)^{10}$

Solution

General term in the expansion of $(x+a)^n$ is $t_{r+1} = nC_r x^{n-r} a^r$ ...(1)

To find the 5th term of $\left(x-\frac{3}{x^2}\right)^{10}$

For this take $r = 4$

Then,

$$t_{4+1} = t_5 = 10C_4 (x)^6 \left(-\frac{3}{x^2}\right)^4$$

(Here $n = 10$, $x = x$, $a = -\frac{3}{x^2}$)

$$= 10C_4 (x)^6 \frac{3^4}{x^8}$$

$$= \frac{17010}{x^2}$$

Example 2.43

Find the middle term in the expansion of $\left(x^2-\frac{2}{x}\right)^{10}$

Solution

Compare $\left(x^2-\frac{2}{x}\right)^{10}$ with $(x+a)^n$

$n = 10$, $x = x^2$, $a = -\frac{2}{x}$

Since $n = 10$, we have 11 terms (odd)

$\therefore$ 6th term is the middle term.

The general term is

$t_{r+1} = nC_r x^{n-r} a^r$ ...(1)

To get t_6, put $r = 5$

$$t_{5+1} = t_6 = 10C_5 (x^2)^5 \left(\frac{-2}{x}\right)^5$$

$$= 10C_5 x^{10} \frac{(-2)^5}{x^5}$$

$$= -8064x^5$$

Example 2.44

Find the middle term in the expansion of $\left(\frac{x}{3}+9y\right)^9$

Solution

Compare $\left(\frac{x}{3}+9y\right)^9$ with $(x+a)^n$

Since $n = 9$,we have 10 terms(even)

$\therefore$ There are two middle terms namely $\frac{t_{n+1}}{2}$, $\frac{t_{n+3}}{2}$ i.e., $\frac{t_{9+1}}{2}$, $\frac{t_{9+3}}{2}$

General term in the expansion of $(x+a)^n$ is

$t_{r+1} = nC_r x^{n-r} a^r$ ... (1)

Here t_5 and t_6 are middle terms.

put $r = 4$ in (1),

$$t_{4+1} = t_5 = 9C_4 \left(\frac{x}{3}\right)^5 \cdot (9y)^4$$

$$= 9C_4 \frac{x^5}{3^5} \cdot 9^4 y^4$$

$$= \frac{9\times8\times7\times6}{4\times3\times2\times1} \cdot \frac{x^5}{3^5} \cdot 9^4 y^4$$

$$= 3402\, x^5 y^4$$

put $r = 5$ in (1),

$$t_{5+1} = t_6 = 9C_5 \left(\frac{x}{3}\right)^4 \cdot (9y)^5$$

$$= 9C_5 \frac{x^4}{3^4} \cdot 9^5 y^5$$

$$= 91854\, x^4 y^5$$

Example 2.45

Find the Coefficient of x^{10} in the binomial expansion of $\left(2x^2-\frac{3}{x}\right)^{11}$

Solution

General term of $(x+a)^n$ is

$t_{r+1} = nC_r x^{n-r} a^r$ (1)

Compare $\left(2x^2-\frac{3}{x}\right)^{11}$ with $(x+a)^n$

02_11th_BM-STAT_Ch-2-EM.indd 44 21-04-2020 12:18:59 PM